Experiment 17: Accessing WordNet Synsets and Meanings in Python

1. Aim

To write a Python program that demonstrates how to access WordNet to retrieve synsets and explore word meanings.

2. Procedure

1. Import `wordnet` from the `nltk.corpus` module.
2. Use `wordnet.synsets()` to retrieve all synsets for a given word.
3. Access the first synset and extract its definition using `definition()`.
4. Extract the synonyms (lemma names) using `lemma_names()`.
5. Display the synset list, definition, and synonyms.

3. Program

```
from nltk.corpus import wordnet

syns = wordnet.synsets("program")
print("Synset Name:", syns[0].name())

print("Definition:", syns[0].definition())

print("Synonyms:", syns[0].lemma_names())
```

4. Output

```
Synset Name: plan.n.01
Definition: a series of steps to be carried out or goals to be
accomplished
Synonyms: ['plan', 'program', 'programme']
```

5. Result

Thus, the Python program was successfully executed to access WordNet, retrieve synsets, and explore word meanings using `wordnet.synsets()`.